

The Perrone–Erdős Ellipsoid

Using the Erdős Primitive Set Bound (Lichtman 2022) as Inertia Ellipsoid Parameter and Imaginary Coefficient

Extended with Adelic Framework and Formal Proof Strategy

Background and Attribution

The constant $\beta \approx 1.6366$ is not original to this work. It is the universal upper bound established by the **Erdős primitive set conjecture (#164)** proven by Lichtman, which states that for any primitive set A , the sum

$$\sum_{n \in A} 1/(n \log n)$$

is strictly maximized when A is the set of all primes. This conjecture was **proved by Jared Duker Lichtman in 2022** (*Annals of Mathematics*, 2023). The prime value of the bound is:

$$\beta := \sum_{p \text{ prime}} 1/(p \log p) \approx 1.6366243912\dots$$

Original contribution of this work: the observation that β can serve simultaneously as:

- the semi-axis parameter $b = \sqrt{\beta}$ of an inertia ellipsoid E_β whose arc-length extrema align with the prime numbers
- the imaginary coefficient in a complex tension function $T(n)$ defined on the Perrone ellipse
- a spectral probe of the Riemann Hypothesis via its convergence pattern
- an adelic trace $\text{Tr}(\Phi \mid L^2(\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}))$ connecting primes and Riemann zeros through Hecke duality

No prior work known to the author has proposed this geometric and spectral reuse of β . The combination is referred to here as the **Perrone–Erdős construction**.

1. The Erdős–Lichtman Constant β

1.1 Definition (Primitive Sets)

A set $A \subset \mathbb{N}_{\geq 1}$ is *primitive* if no element divides another.

Theorem (Lichtman 2022): For any primitive set A ,

$$\sum_{n \in A} 1/(n \log n) \leq \sum_{p \text{ prime}} 1/(p \log p) =: \beta$$

with equality if and only if $A = \mathbb{P}$ (the primes).

The weight

$$1/(n \log n)$$

equals the bi-harmonic density

$$v_0(n) = d/dn(\log \log n)$$

The convergence of

$$\sum_p 1/(p \log p)$$

follows from the extra $\log p$ in the denominator.

1.2 Numerical Value

Primes considered	Partial β
First 100 primes (up to 541)	1.47838998
First 1,000 primes (up to 7,919)	1.52534131
First 10,000 primes (up to 104,729)	1.55012678
First 100,000 primes (up to 1,299,709)	1.56423623
Tail estimate $\sum_{p>10^6}$	+0.07238
Full β	≈ 1.6366243912

Convergence rate is

$$O(1/\log N)$$

reflecting the rate at which the prime distribution fills the bi-harmonic measure.

1.3 Analytic Structure

Decomposition

$$\beta = 1/\log(2) + \Delta$$

- $1/\log(2) \approx 1.4427$: smooth contribution
- $\Delta \approx 0.1939$: excess from non-uniformity of the primes

Integral representation

Via the prime zeta function:

$$\beta = \int_1^\infty P(u) \, du$$

and

$$\beta = \sum_{n=1}^\infty \mu(n)/n \cdot \int_1^\infty \log \zeta(nu) \, du$$

2. The Perrone–Erdős Inertia Ellipsoid

The **Perrone–Erdős ellipsoid** is defined by:

$$E_\beta : x^2 + y^2/\beta = 1$$

with semi-axes:

- $a = 1$
- $b = \sqrt{\beta} \approx 1.2793$

and eccentricity:

$$e = \sqrt{1 - 1/\beta} \approx 0.6237$$

2.1 The Arc-Length Tension Function

Parameterization:

$$P(\theta) = (\cos \theta, \sqrt{\beta} \cdot \sin \theta)$$

Arc-length element:

$$ds = \sqrt{(\sin^2 \theta + \beta \cos^2 \theta)} \, d\theta$$

Define:

$$\theta = \log n$$

Then:

$$T(n) = \sqrt{(\sin^2(\log n) + \beta \cos^2(\log n))} / \log n$$

Extrema occur at:

$$\theta = k\pi/2$$

thus:

$$n_k = e^{\{k\pi/2\}}$$

k	θ	n_k	$T(n_k)$	Type	Nearest prime
1	1.5708	4.81	0.6366	min	5
2	3.1416	23.14	0.4072	max	23
3	4.7124	111.3	0.2122	min	109
4	6.2832	535.5	0.2036	max	541
5	7.8540	2576.0	0.1273	min	2579
6	9.4248	12391.6	0.1357	max	12391
7	10.9956	59609.7	0.0909	min	59611

2.2 β as Imaginary Coefficient

Define the complex tension function:

$$T\mathbb{C}(n) = [\sin(\log n) + i\beta \cos(\log n)] / \log n$$

The modulus satisfies:

$$|T\mathbb{C}(n)| = T(n)$$

and the argument is:

$$\arg(T\mathbb{C}(n)) = \arctan(\beta \cot(\log n))$$

3. The Period Connection

The period of $T(n)$ in log-space is:

$$2\pi \approx 6.283$$

with quarter-period:

$$\pi/2 \approx 1.571$$

The empirical period:

$$\Omega \approx 4.911$$

is close to:

$$3\pi/2 \approx 4.712$$

Conjecture (Perrone Period)

$$\int_2^{e^\Omega} v_0\text{primes}(t) dt = (1/4)\beta$$

4. The Resonance Function and Explicit Formula

4.1 Derivation from $\psi(x)$

Under RH:

$$\psi(x) = x - \sum_{\rho} x^{\rho/\rho} - \log(2\pi) + O(x^{-1})$$

Define the resonance function:

$$R(x) = \sum_{\gamma} \cos(\gamma \log x) / (1/4 + \gamma^2)$$

4.2 The Bi-Harmonic Weight as Spectral Density

$$\hat{v}_0(\gamma) = \int_0^\infty (e^{-t}/t) e^{-i\gamma t} dt \rightarrow 1/(1/4 + \gamma^2)$$

Key insight:

$$v_0(n) = 1/(n \log n)$$

produces the resonance kernel of the explicit formula.

5. The Adelic Framework

5.1 The Formal Gap

The resonance function is defined through zeros of $\zeta(s)$, not directly through primes.

5.2 The Adelic Space

$$\mathbb{A}_{\mathbb{Q}} = \mathbb{R} \times \prod_p \mathbb{Q}_p$$

- primes appear as eigenvalues of Frobenius action on $\mathbb{Q}_p$
- zeros appear as frequencies of a self-adjoint operator on $L^2(\mathbb{A}_{\mathbb{Q}})$
- β becomes an adelic trace quantity

5.3 Three-Step Formal Proof Strategy

Step 1

$$R(x) = \int_{\{\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}\}} v_0(|a|_{\mathbb{A}}) \cdot \psi(ax) \, da$$

Step 2

Adelic Poisson transform yields:

$$1/(1/4 + y^2)$$

Step 3

Localization via p-adic norm implies concentration near primes.

5.4 Conjecture 4 (Adelic)

There exists a self-adjoint operator H on

$$L^2(\mathbb{A}_{\mathbb{Q}}/\mathbb{Q})$$

such that:

- $\text{spec}(H) = \{\gamma : \zeta(1/2+i\gamma)=0\}$
- eigenvalues on $L^2(\mathbb{Q}_p)$ equal $\log p$
- $\beta = \text{Tr}(e^{-H})$

6. Conjectures

Conjecture 1 (β -RH Bridge)

$$\Delta = -2 \sum_{\gamma>0} \text{Re}[\zeta'(1/2+i\gamma)/(\gamma \cdot \zeta(1/2+i\gamma))]$$

Conjecture 2 (Ellipse-Gap Correspondence)

For extrema:

$$n_k = e^{\{k\pi/2\}}$$

nearest primes satisfy:

$$|p_k - n_k| = O(\sqrt{n_k} \cdot \log n_k)$$

Conjecture 3 (β as Zero-Detector)

Residual oscillations in partial sums of β detect off-critical zeros.

Conjecture 4 (Adelic)

Conjectures 1–3 are subsumed by the existence of the operator H .

7. Implementation Priorities

Priority 1: Compute β to 100 digits

```
import mpmath as mp
from sympy import nextprime

mp.dps = 100
beta = mp.mpf(0)
p = 2
```

```

while p < 10**7:
    beta += mp.mpf(1) / (p * mp.log(p))
    p = int(nextprime(p))

beta += mp.mpf(1) / mp.log(10**7)
print(beta)

```

Priority 2

Residual oscillation test using FFT.

Priority 3

Ellipse extrema vs prime gaps.

Priority 4

Compute resonance scans using real Riemann zeros.

Priority 5

Spectral Δ via Möbius representation.

Summary

The Erdős–Lichtman constant

$$\beta = \sum_{p \text{ prime}} 1/(p \log p) \approx 1.6366$$

is:

- a convergent extremal constant for primitive sets
- reused as the squared semi-axis of the Perrone–Erdős ellipsoid
- reused as the imaginary coefficient of a complex tension function
- expressible through the prime zeta function
- decomposable as:

$$1/\log 2 + \Delta$$

- interpretable as an adelic trace quantity
- proposed as a numerical probe of the Riemann Hypothesis

The Perrone–Erdős construction contains four falsifiable conjectures and a three-step adelic proof strategy connecting primes, resonance, and spectral geometry.